Ph429: Nonlinear Dynamics & Chaos

Problem Set 4

Zeshui Song

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Problem 1

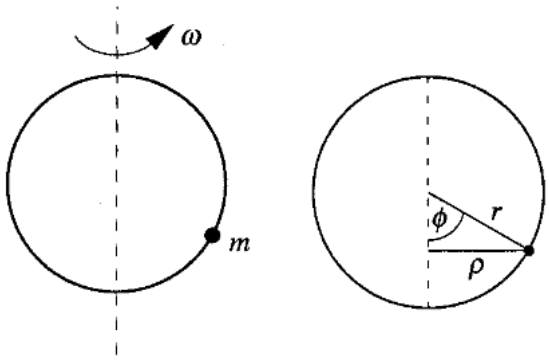
In class we looked at the "Bead on a Hoop, Submerged in Syrup" problem. We discussed the stability of the fixed points:

$$x^* = 0 \qquad \text{top of hoop} \tag{1}$$

$$x^* = \pi \qquad \text{bottom of hoop} \tag{2}$$

$$x^* = \pm \cos^{-1}\left(\frac{g}{r\omega^2}\right) \tag{3}$$

but we did not demonstrate their stability. Find their stability graphically and algebraically. Do your best.



Solution The equation of motion for the bead on a hoop submerged in syrup is given by:

$$mr\ddot{\phi} = -b\dot{\phi} - mg\sin\phi + mr\omega^2\sin\phi\cos\phi$$

For an overdamped system where the viscous damping dominates over inertial effects, we can neglect the inertial term ($\ddot{\phi} \approx 0$), leading to the first order approximation:

$$\dot{\phi} = -\frac{mg}{b}\sin\phi + \frac{mr\omega^2}{b}\sin\phi\cos\phi$$

Test for stability at fixed points: Let $f(\phi) = \dot{\phi}$

$$f'(\phi) = \frac{df}{d\phi} = -\frac{mg}{b}\cos\phi + \frac{mr\omega^2}{b}(\cos^2\phi - \sin^2\phi)$$

- **Fixed point at $\phi^* = 0$:**

$$f'(0) = -\frac{mg}{b} + \frac{mr\omega^2}{b} = \frac{m}{b}(r\omega^2 - g)$$

  - If $r\omega^2 < g$, then $f'(0) < 0$, indicating a stable fixed point.
  - If $r\omega^2 > g$, then $f'(0) > 0$, indicating an unstable fixed point.
  - If $r\omega^2 = g$, then $f'(0) = 0$, which is inconclusive.

- **Fixed point at $\phi^* = \pi$:**

$$f'(\pi) = \frac{mg}{b} + \frac{mr\omega^2}{b} = \frac{m}{b}(g + r\omega^2)$$

  Since $f'(\pi) > 0$, this fixed point is always unstable.

- **Fixed points at $\phi^* = \pm\cos^{-1}\left(\frac{g}{r\omega^2}\right)$:**

  Using the identity $\sin^2\phi = 1 - \cos^2\phi$, rewrite $f'(\phi)$ as:

$$f'(\phi) = -\frac{mg}{b}\cos\phi + \frac{mr\omega^2}{b}(2\cos^2\phi - 1)$$

$$\begin{aligned} f'\left(\pm\cos^{-1}\left(\frac{g}{r\omega^2}\right)\right) &= -\frac{mg}{b}\left(\frac{g}{r\omega^2}\right) + \frac{mr\omega^2}{b}\left(2\left(\frac{g}{r\omega^2}\right)^2 - 1\right) \\ &= \frac{m}{b}\left(-\frac{g^2}{r\omega^2} + \frac{2g^2}{r\omega^2} - r\omega^2\right) \\ &= \frac{m}{b}\left(\frac{g^2}{r\omega^2} - r\omega^2\right) \\ &= \frac{m}{br\omega^2}\left(g^2 - r^2\omega^4\right) \end{aligned}$$

  - If $g^2 > r^2\omega^4$, then $\phi^* = \pm\cos^{-1}\left(\frac{g}{r\omega^2}\right)$, do not exist since $\cos^{-1}$ is only defined for values in the range $[-1, 1]$.
  - If $g^2 < r^2\omega^4$, then $f'\left(\pm\cos^{-1}\left(\frac{g}{r\omega^2}\right)\right) < 0$, indicating stable fixed points.
  - If $g^2 = r^2\omega^4$, then $f'\left(\pm\cos^{-1}\left(\frac{g}{r\omega^2}\right)\right) = 0$, which is inconclusive.

Graphically testing for stability:

Since in the stability analysis, the relationship between $r\omega^2$ and g is important, I will define the parameter

$$\lambda = \frac{r\omega^2}{g}$$

such that

$$\dot{\phi} = \frac{mg}{b}\sin\phi\left(\frac{r\omega^2}{g}\cos\phi - 1\right) = \frac{mg}{b}\sin\phi(\lambda\cos\phi - 1)$$

Rescale time by defining $\tau = \frac{mg}{b}t$, then we have

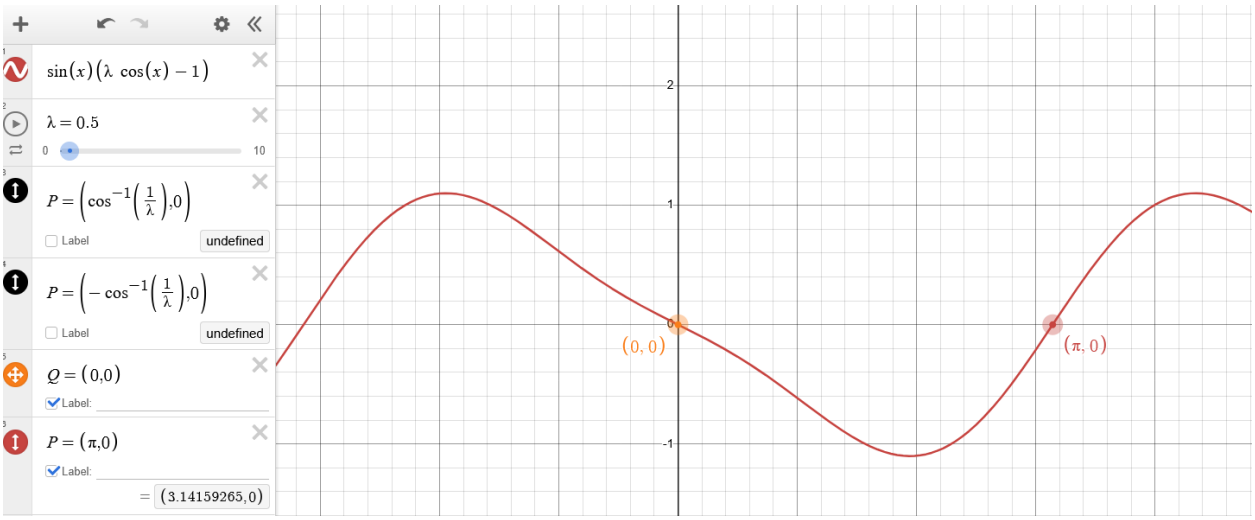
$$\frac{d\phi}{dt} = \frac{d\phi}{d\tau}\frac{d\tau}{dt} = \frac{d\phi}{d\tau}\cdot\frac{mg}{b}$$

Using the rescaled time, we have the following first order ODE as a function of λ and ϕ:

$$\frac{d\phi}{d\tau} = \sin\phi(\lambda\cos\phi - 1)$$

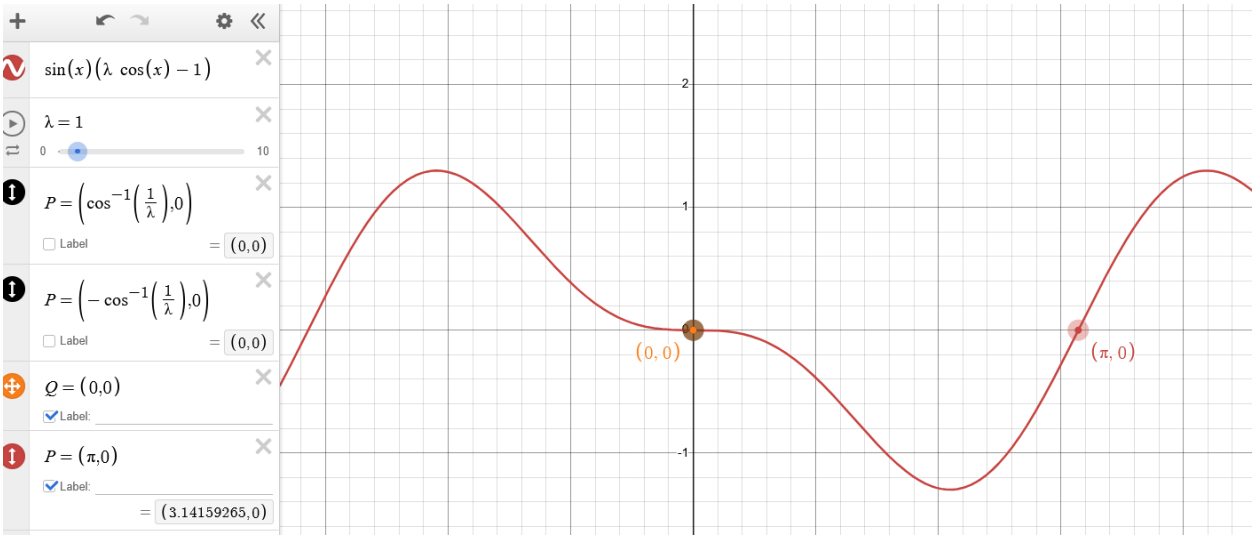
Plotting the phase portrait of $\frac{d\phi}{d\tau}$ vs ϕ for different values of λ will show the stability of the fixed points.

- $0 < \lambda < 1$: $g > r\omega^2$



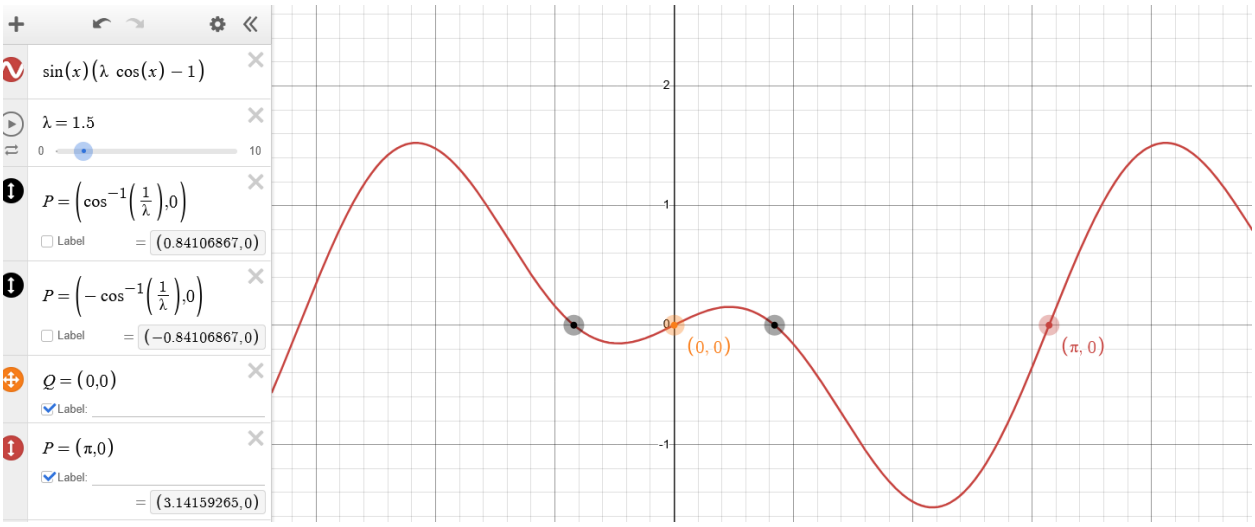
$\phi^* = 0$ is stable, $\phi^* = \pi$ is unstable, and $\phi^* = \pm\cos^{-1}\left(\frac{g}{r\omega^2}\right)$ do not exist.

- $\lambda = 1$: $g = r\omega^2$



$\phi^* = 0$ is stable, $\phi^* = \pi$ is unstable, and $\phi^* = \pm\cos^{-1}\left(\frac{g}{r\omega^2}\right)$ coincides with $\phi^* = 0$.

- $\lambda > 1$: $g < r\omega^2$

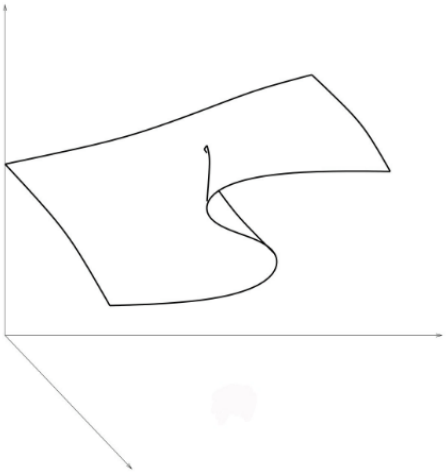


$\phi^* = -\cos^{-1}\left(\frac{g}{r\omega^2}\right)$ is stable, $\phi^* = 0$ is unstable, $\phi^* = \cos^{-1}\left(\frac{g}{r\omega^2}\right)$ is stable, and $\phi^* = \pi$ is unstable.

Problem 2

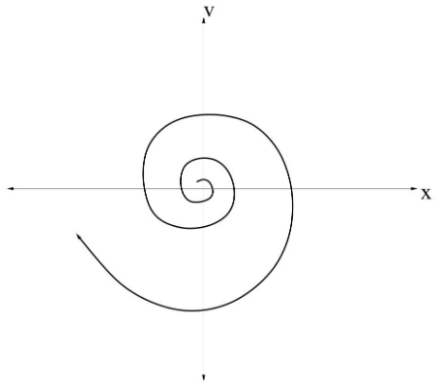
For the following two figures, please invent a system that could produce them, giving it both in mathematical form and as a model of something (be creative, tell me a story).

(a)



Solution

(b)



Solution